

Checkout Experiment & Growth Analytics – Final Memo

Business Objective:

The objective of this project was to evaluate the performance of a newly introduced checkout experience (Variant B) in an e-commerce platform. The company aimed to reduce checkout drop-offs and increase completed purchases by simplifying the checkout process. The analysis investigates whether Variant B should be rolled out to all users, identifies the largest funnel drop-offs, examines performance across user segments, and estimates the business impact of a full rollout over the next 30 days.

Data & Methodology:

Multiple datasets were integrated and processed through a Python ETL pipeline. Raw data sources included users, sessions, events, orders, order items, product catalog data, and marketing campaign metadata. Data cleaning steps included removing duplicates, standardizing categorical values, handling missing fields, and identifying potential outliers in revenue values. Three curated analytical datasets were created: fact_sessions (session-level data), fact_orders (order-level data), and dim_users_enriched (user-level metrics). These datasets enabled funnel reconstruction, experiment comparison, and segment-level analysis.

Core Business Metrics:

Metric Value Total Revenue \$5.03M Total Orders 705 Average Order Value (AOV) \$7.13K Conversion Rate 7.84% These metrics establish the baseline performance of the e-commerce platform during the analysis period. The revenue and order volume indicate strong purchasing activity, while the conversion rate suggests there is still opportunity to reduce friction in the purchase journey.

Funnel Analysis:

Reconstruction of the user funnel revealed that the largest drop-off occurs between 'Add to Cart' and 'Begin Checkout'. This stage represents a critical transition from purchase intent to transaction commitment. High drop-off at this point suggests that users may encounter friction before starting the checkout process. Possible causes include unclear pricing, shipping cost visibility, or complex checkout entry steps. Improving the transition into checkout can therefore produce meaningful gains in overall conversion.

A/B Experiment Results:

The experiment compared the performance of Variant A (existing checkout flow) and Variant B (new checkout design). The results indicate a clear improvement in conversion under the new design. Variant Conversion Rate Variant A 8.38% Variant B 9.99% Variant B improves conversion by approximately 1.61 percentage points, which corresponds to a relative improvement of roughly 19%. This uplift suggests that the redesigned checkout reduces friction and helps more users complete their purchases.

Estimated Business Impact:

Using the observed experiment uplift and the current average order value, the expected impact of rolling out Variant B across all sessions was estimated. If the improved conversion rate is sustained, the platform could generate approximately 145 additional completed orders over the next 30 days. Based on the current average order value, this would translate to roughly \$1.03M in additional monthly revenue.

Recommendation:

Based on the experiment results and supporting analysis, the recommended action is to roll out Variant B to all users. The observed improvement in conversion rate indicates

that the new checkout design meaningfully reduces friction during the purchase process. Following rollout, the company should continue monitoring key performance indicators and run additional experiments to further optimize checkout performance.

Risks & Next Steps:

While the results are promising, several factors should be monitored after deployment. Seasonal demand changes, traffic mix shifts, or segment-specific behavior could affect observed performance. Continued A/B testing and monitoring of conversion metrics will help ensure the improvement remains consistent. Future experiments may focus on mobile checkout optimization, payment flow simplification, and reducing friction earlier in the purchase journey.